Experiment 4

Title: Implementation of N-Gram (Bi-Gram) Model

Aim

To implement the **N-Gram model (Bi-Gram)** for analyzing word sequences in a given text dataset.

Objectives

1. To understand the concept of the N-Gram model in text processing.
2. To study the working of the Bi-Gram model for word sequence analysis.
3. To implement the Bi-Gram technique on a given text.

Theory

1. Introduction to N-Gram Model
The N-Gram model is a technique used in **Natural Language Processing (NLP)** to analyze sequences of words in a text. It predicts the probability of a word based on the previous words in the sequence. N-Grams help in understanding patterns and relationships between words in a sentence.

2. Types of N-Grams
N-Grams are classified based on the number of words considered together. A **unigram** consists of a single word, a **bigram** consists of two consecutive words, and a **trigram** consists of three consecutive words. These models are useful for studying language patterns and predicting text sequences.

3. Bi-Gram Model
A Bi-Gram model is a type of N-Gram where **two consecutive words are considered as a pair**. For example, in the sentence "Data science is useful", the bigrams would be *data science*, *science is*, and *is useful*. This model helps in understanding the relationship between adjacent words.

4. Applications of Bi-Gram Model
Bi-Gram models are widely used in **language modeling, speech recognition, text prediction, and machine translation**. They help computers understand word relationships and improve the accuracy of many NLP systems.

Conclusion

In this experiment, the concept and implementation of the Bi-Gram model were studied. The model helps analyze word sequences by considering pairs of consecutive words. This approach makes it easier to identify language patterns and relationships between words. Bi-Gram models are widely used in many NLP applications. Thus, they play an important role in text analysis and language modeling.